

Bias-Variance Tradeoff & Overfitting vs. Underfitting

A conceptual bridge to Regularization

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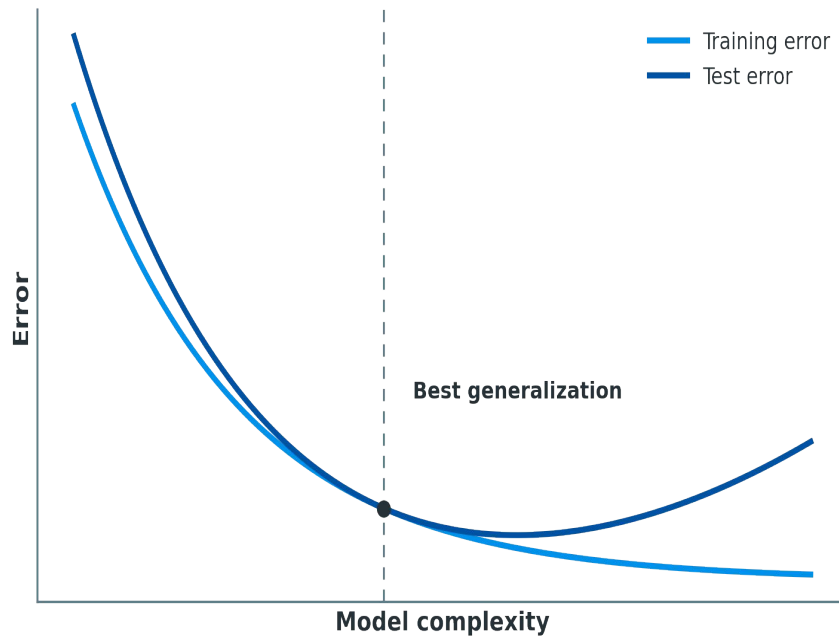
1.

Part 1

Linear Regression

Overfitting vs. Underfitting

- Underfitting: model too simple, misses the true pattern
- Overfitting: model too complex, memorizes training noise
- Training error always falls as complexity increases
- Test error follows a U-shape — the real target to minimize

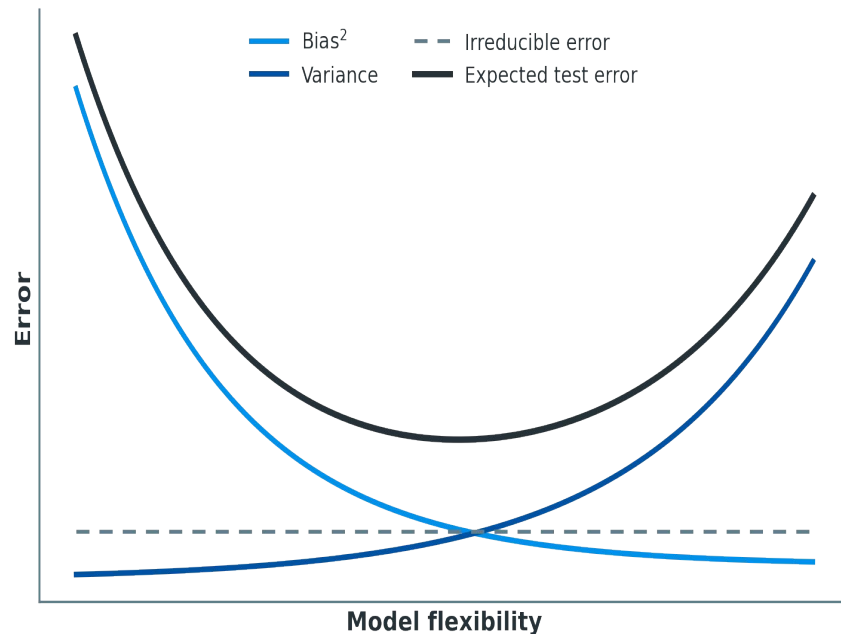


Plot: Training error vs. Test error across model complexity

The Bias-Variance Decomposition

- Expected test error splits into three components
- Bias: error from overly simple assumptions
- Variance: sensitivity to fluctuations in training data
- Irreducible error: noise that no model can remove

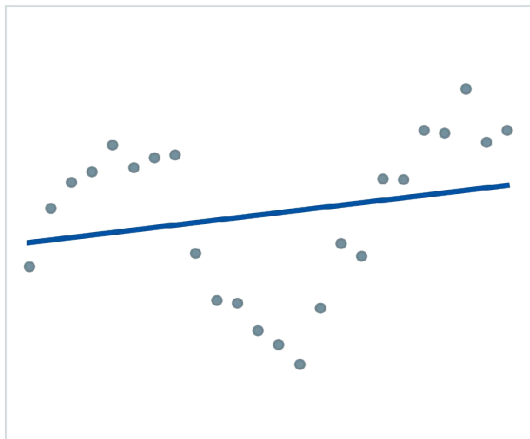
$$E[(y_0 - f(x_0))^2] = \text{Var}(f(x_0)) + [\text{Bias}(f(x_0))]^2 + \text{Var}(\varepsilon)$$



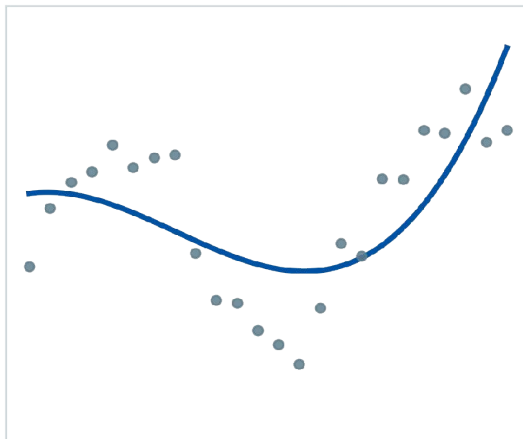
Plot: Bias², variance, irreducible error and their sum vs. flexibility

Model Flexibility and Fit

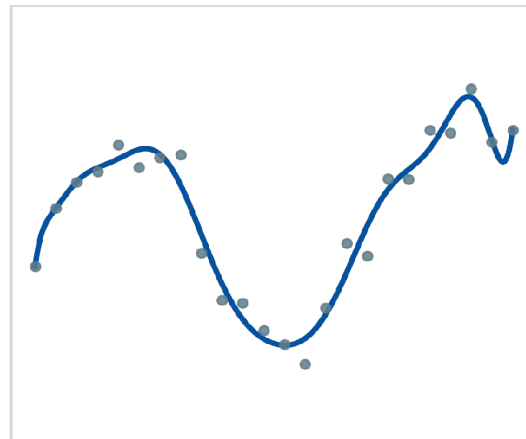
**Underfit
(high bias)**



**Good fit
(balanced)**



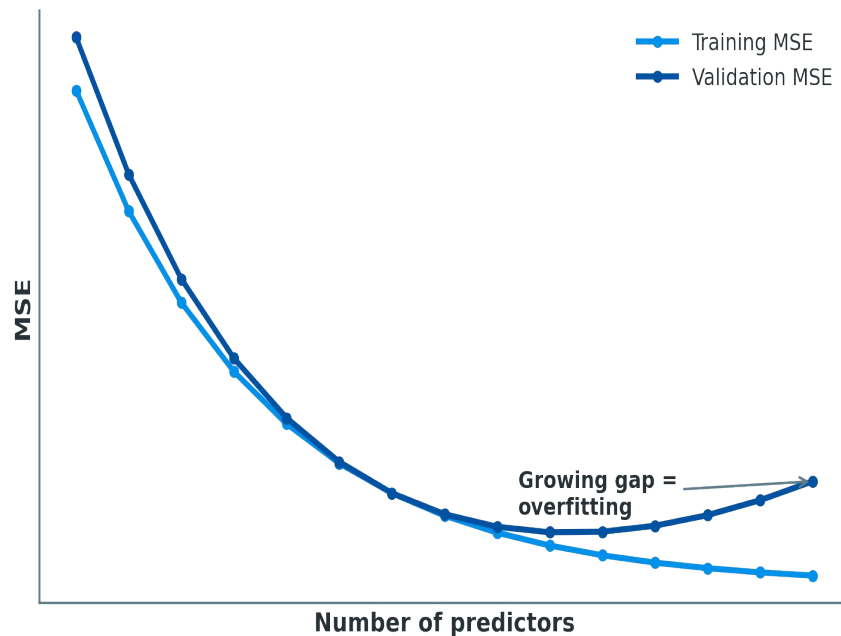
**Overfit
(high variance)**



- Simple models (few predictors): high bias, low variance
- Flexible models (many predictors, high-order terms): low bias, high variance
- The goal is the flexibility that minimizes total test error, not training error

Detecting Overfitting in Linear Regression

- Training R^2 keeps rising as predictors are added
- Validation MSE stops improving, then increases
- A growing train-validation gap signals overfitting
- Irrelevant predictors inflate variance without adding signal



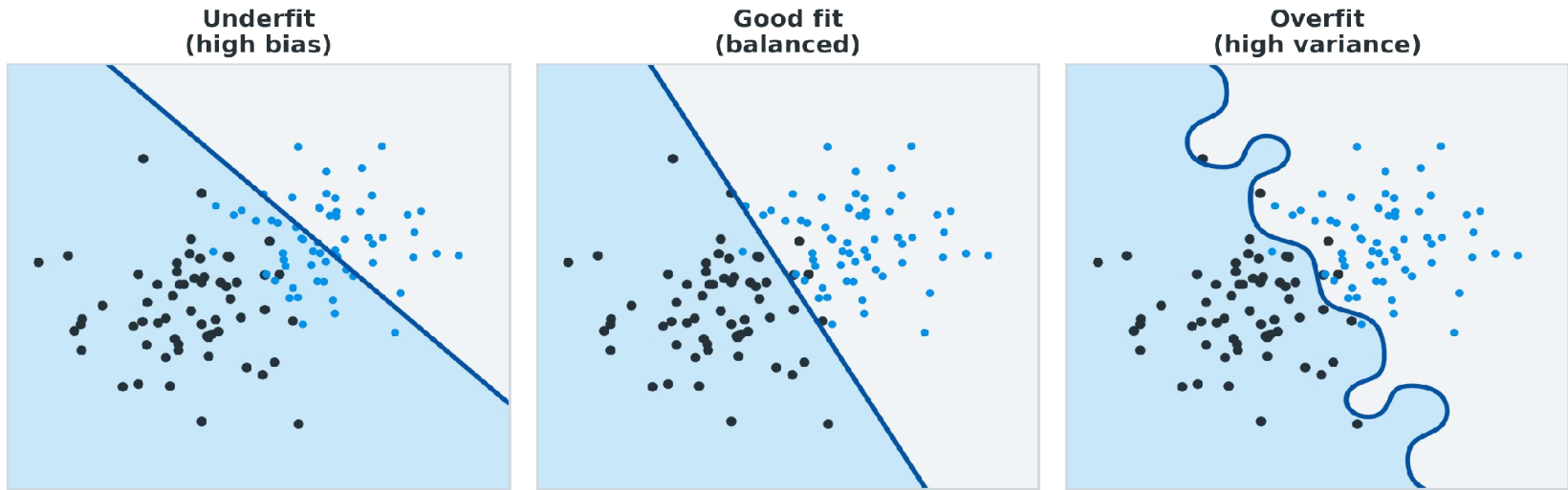
Plot: Training MSE vs. Validation MSE as predictors are added

2.

Part 2

Logistic Regression

Overfitting vs. Underfitting in Classification

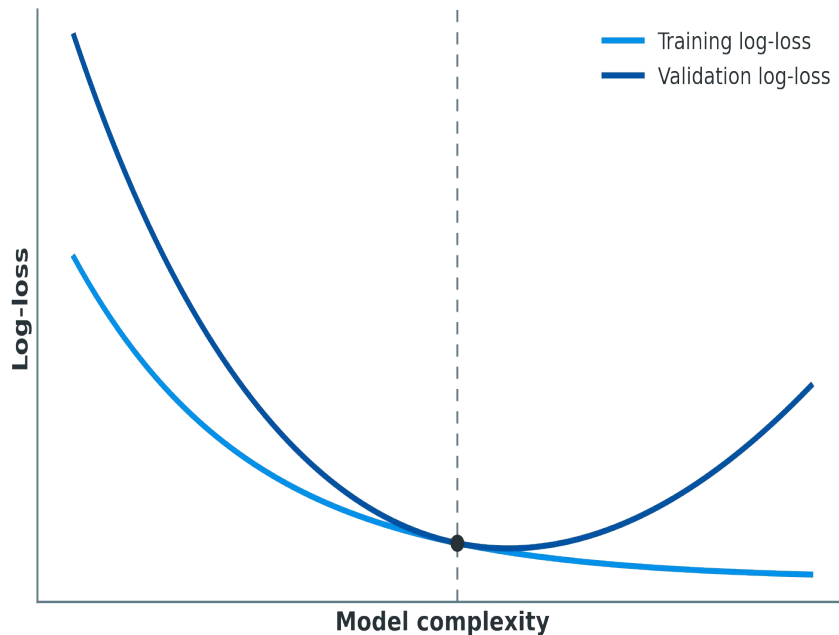


- Underfit: boundary too simple, poor accuracy everywhere
- Overfit: boundary too complex, memorizes individual points
- Assess via training vs. validation accuracy and log-loss

Bias-Variance Tradeoff in Logistic Regression

- Same tradeoff, now measured on log-loss / classification error
- Too few predictors or interactions → high bias (underfit)
- Too many predictors or high-order terms → high variance (overfit)
- Complexity here means predictors, polynomial terms, interactions

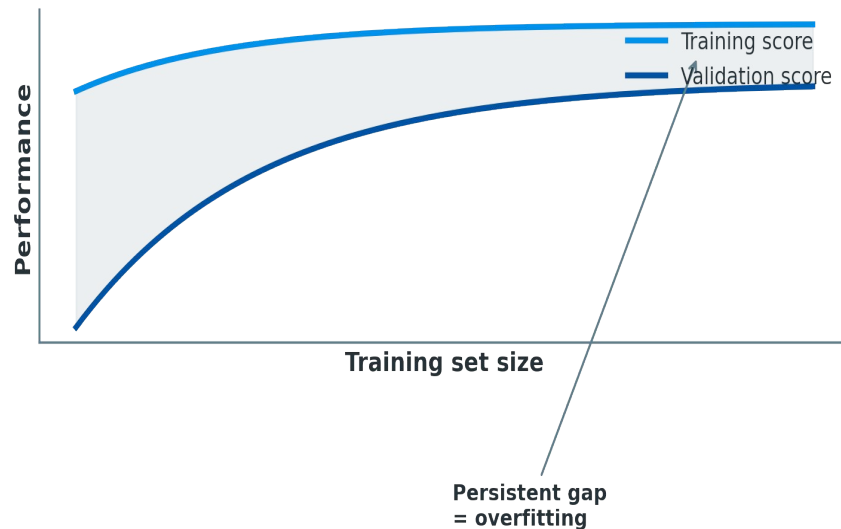
Expected test log-loss $\approx$ Bias + Variance + Irreducible error



Plot: Training vs. Validation log-loss across model complexity

Diagnosing Overfitting: Learning Curves

- Plot training vs. validation performance vs. training set size
- Curves converging near a strong score = good fit
- A large, persistent gap = overfitting
- Both curves plateau low = underfitting



Plot: Training vs. Validation score as training set size grows

Linear vs. Logistic: Same Tradeoff, Different Loss

	Linear Regression	Logistic Regression
Target	Continuous response	Class probability
Loss minimized	Mean squared error	Log-loss (cross-entropy)
Fitting method	OLS / gradient descent	MLE via gradient methods
Complexity driver	Predictors, polynomial terms	Predictors, interaction terms
Regularized by	Ridge / Lasso on β	Ridge / Lasso on β

Why This Matters for Regularization

- Both models face the same bias-variance tradeoff
- Regularization (L1 / L2) directly trades bias for variance
- Ridge shrinks coefficients smoothly toward zero
- Lasso can shrink coefficients exactly to zero — built-in feature selection

Next: Ridge Regression and The Lasso

Thanks!

Any questions?